Index

Name : ..... Class : ..... Year : .....

Expt. No.	Date	Title of the Experiments	Page No.	Date of Submission	Remarks
1.		Resistivity of wire	1-3		
2.		Meter bridge - Resistance of given wire	4-6		
3.		Meter bridge - combination of Resistance	7-10		
4.		Figure of merit	11-13		
5.		Angle of minimum Deviation	14-17		
6.		Refractive index of glass	18-20		
7.		Focal length of convex lens	21-23		
8.		Refractive Index of water by convex lens	24-25		
9.		Semi-conductor diode	26-27		

Scanned with OKEN Scanner

Index

Name : ..... Class : ..... Year : .....

Expt. No.	Date	Title of the Experiments	Page No.	Date of Submission	Remarks
1.		Activity - 1	28-30		
2.		Activity - 2	31		
3.		Activity - 3	32-33		
4.		Activity - 4	34-36		
5		Activity - 5	37-39		
6		Activity - 6	40-42		

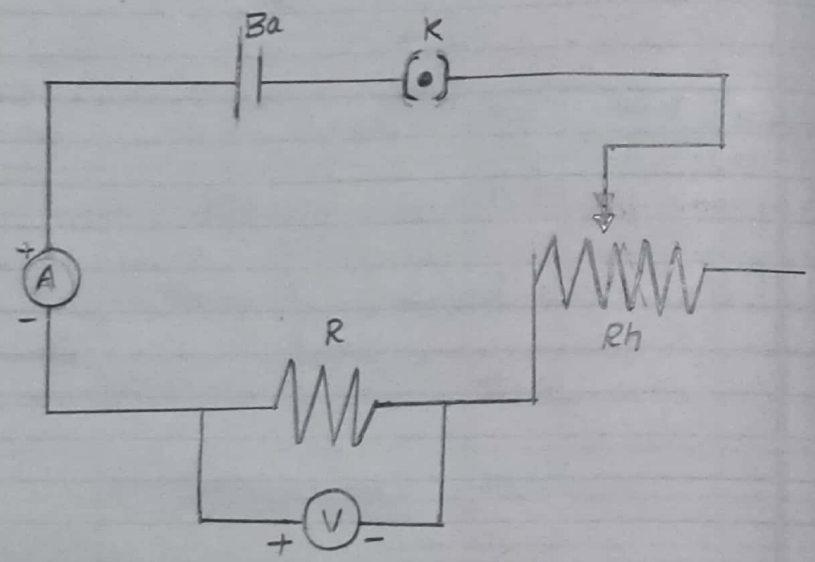
Expt. No.: .....

Date .....

Page No. .....

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Circuit Diagram :-

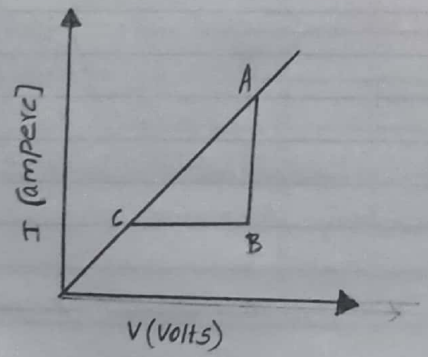


Where:

Ba - Battery, K - plug key, A - Ammeter, V - Voltmeter,

Rh - Rheostat, R - Resistance of Given wire.

Nature of Graph :-



RESISTIVITY OF WIRE

Expt. No.: 1 Date :

Page No.: 07

Aim :-

To determine resistivity of a given wire by plotting graph of potential difference versus current.

Apparatus :-

*One wire of unknown resistance
*Battery Eliminator (0-3V)
*Voltmeter (0-5V)
*Ammeter (0-5A)
*Rheostat
*Plug key
*Connecting wires.

Formula :-

$$R = \frac{1}{\text{slope}} \Omega$$

Resistivity $\rho = \frac{RA}{L}$

Procedure :-

1) Connect various components - resistance wire, rheostat, battery, key, voltmeter & ammeter as shown in circuit diagram.

2) Note whether pointers in ammeter and voltmeter coincide with the zero mark on

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Observations:-

Radius of given wire (r) =

Length of the given wire (l) = _____ m

S.NO	Applied potential difference (V in volts)	Current through the wire (I in ampere)

Expt. No.: .....

Date : .....

Page No.: 09

the measuring scale. If it is not so, adjust the pointer

③ Insert the key K and slide the rheostat contact to one of its extreme ends, so that current passing through the resistance wire is minimum.

④ Note the ammeter and voltmeter readings.

⑤ Remove the key K and allow the wire to cool, if heated. Again insert the key. shift the rheostat contact slightly to increase the applied voltage. Note the ammeter and voltmeter reading.

⑥ Repeat step 5 for five to six different settings of the rheostat. Record your observations in a tabular column.

⑦ Plot a graph between the potential difference across the wire (V) V/s current (I).

⑧ The resistance of the given wire is equal to the reciprocal of the slope.

⑨ Resistivity can be calculated by given formulae.

⑩ We will repeat the procedure with one/two more wires.

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calculation:-

From graph slope = $\frac{AB}{BC}$ =

The Resistance of the given wire is

$$R = \frac{1}{\text{slope}} = _____ \Omega$$

Expt. No.: .....

Date : .....

Page No.: 09

Result:-

Resistivity of given wire = _____ Ω-m

Precautions:-

1. The connection should be neat, clean, tight.
2. The rheostat should be moved smoothly.
3. Readings should be taken without any Parallax error.

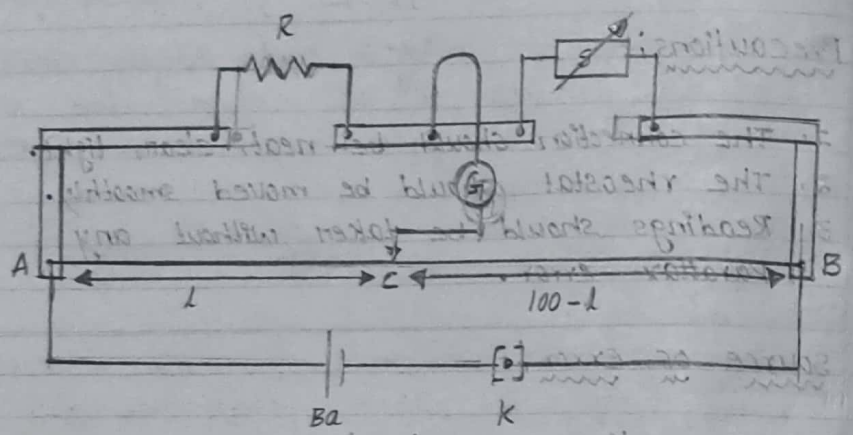
Source of error:-

1. Connections may be loose.
2. The wire may not have uniform thickness.

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Circuit Diagram:-



Where:

AB- Meter bridge wire

Ba – Battery

K – plug key

S - Standard resistance box

R - Resistance of given wire

C - Balancing Point

G – Pointer galvanometer

AC – Balancing length.

METER BRIDGE - RESISTANCE OF GIVEN WIRE

Expt. No.: 2. Date : .....

Page No.: 04

Aim:-

To determine the resistance of the given wire using meter bridge and hence determine the resistivity of the material of the wire.

Apparatus:-

* Meter Bridge
* Experimental wire
* Battery
* Plug Key
* Resistance box
* Jockey
* Pointer galvanometer

Principle:-

A meter bridge works on the Principle of balanced wheatstone's bridge. The wheatstone bridge is said to be balanced when the current through the galvanometer is zero.

Formula:-

Resistance of the wire

$$R = \frac{Sl}{100 - l} \ \Omega$$

where:

S – standard resistance.

Teacher's Signature: .....

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Observations;-

Length of the given wire (L) = ~~_____~~ Cm

Radius of the given wire (r) = _____ Cm

TO Measure the resistance of the given wire:-

S.NO	Resistance S in Ω	Balancing length l in cm	$R = \frac{Sl}{100-l}$ in Ω

Mean R = _____ Ω

Expt. No.: .....

Date : .....

Page No.: 05

l - Balancing length.

L - Length of the experimental wire.

r - Radius of the experimental wire.

Procedure:-

1. The connections are made as shown in the circuit diagram.

2. A suitable resistance S is unplugged in the standard resistance box.

3. The circuit is checked for opposite deflections by Placing jockey at the two ends of the wire AB alternately.

4. The jockey is moved on the wire from the end A towards B till the galvanometer shows zero deflection.

5. The balancing length l is measured. The resistance of the wire is calculated using the formula

$$R = \frac{Sl}{100-l}$$

6. The experiment is repeated for different values of S and average value of R is found.

Teacher's Signature: .....

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Expt. No.: .....

Date : .....

Page No.: 06

7. The length and radius of the experimental wire is measured using meter scale and screw guage respectively.

Result :-

The resistance of the given wire is _____ Ω

Precautions :-

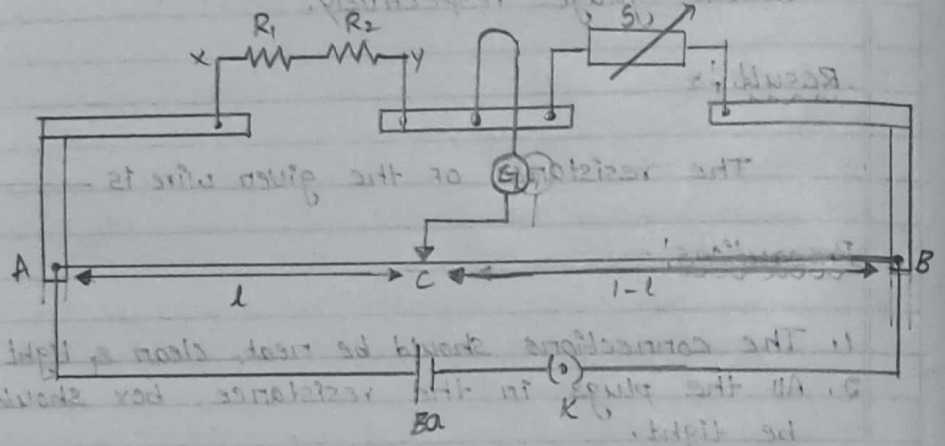
1. The connections should be neat, clean & tight.
2. All the plugs in the resistance box should be tight.
3. Move the jockey gently over the bridge wire and do not rub it.
4. The plug in key K should be inserted only when the observation to be taken.

Source of error :-

1. Connections may be loose.
2. The wire may not have uniform thickness.

Teacher's Signature: .....

Circuit Diagram:-



Where;

R_1 & R_2 – given resistors

S – standard resistance box

G_L – Galvanometer

Ba – Battery

K – Plug key

AB – Meter Bridge wire

AC – Balancing length.

METER BRIDGE – COMBINATION OF RESISTANCES

Expt. No.: 3 Date : .....

Page No.: 08

Aim:-

To verify the laws of combination of resistance in series and using meter bridge.

Apparatus:-

* Meter Bridge
* Battery
* Two resistor
* Plug key
* Resistance box
* Galvanometer
* jockey

Principle:-

The series or Parallel combinations of two resistors are connected in the left gap of Meter Bridge R. A standard resistance S in connected in the right gap of the meter bridge. When the bridge is balanced the current through the galvanometer is zero. (the jockey is at the Point C). The balancing condition is,

$$\frac{\text{Effective resistance of combination}}{S} = \frac{\text{Resistance of the wire A}}{\text{Resistance of the wire B}}$$

$$= \frac{l}{100-l}$$

$(\because R \propto l)$

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Observation:-

1. Resistance of given resistor, R_1 = _____ Ω

2. Resistance of given resistor, R_2 = _____ Ω

To verify the laws of combination of resistance in series.

S.NO	Resistance S in Ω	Balancing length l in m	$R_s = \frac{Sl}{100-l}$ in Ω
1.			
2.			
3.			

Mean R_s = _____ Ω

Calculation:-

Theoretical, $R_s = R_1 + R_2$

The effective resistance of combination is given by

$$R_s \text{ or } R_p = S\left(\frac{l}{100-l}\right)$$

Formula:-

Resistance

$$R = \frac{Sl}{100-l} \ \Omega$$

Equivalent resistance in series, $R_s = R_1 + R_2$ Ω

where:

S - standard resistance.

l - Balancing length.

R_1 & R_2 - Individual resistances.

Procedure:-

1. The connections are made as shown in the circuit diagram, i.e, the two resistors are connected in series.

2. A suitable resistance S in unplugged in the standard resistance box.

3. The circuit is checked for opposite deflections by placing jockey at the two ends of the meter bridge wire AB alternately.

Teacher's Signature: .....

Scanned with OKEN Scanner

Expt. No.: .....

Date : .....

Page No.: 9

4. The jockey is moved on the wire from the end A towards B till the galvanometer shows zero deflection

5. The balancing length l is measured. The equivalent resistance of the series combination is calculated using the formula

$$R_s = \frac{sl}{100-l}.$$

6. The experiment is repeated for different values of S and average value of R_s is found. The theoretical value of equivalent resistance is calculated using the formula

$$R_s = R_1 + R_2.$$

7. Now the two resistors are connected in parallel on the left gap of Meter Bridge.

8. The jockey is moved on the wire and hence balancing length l is measured. The equivalent resistance of the parallel combination is calculated using the formula

$$R_s = \frac{sl}{100-l}.$$

Teacher's Signature: .....

Expt. No.: .....

Date : .....

Page No.: 10

Result:-

Theoretical value of resistances in series, R_S = ____ Ω

Experimental value of resistances in series, R_S = ____ Ω

Experimental values are in good agreement with theoretical values, thus the laws of combination of resistances in series are verified.

Precautions:-

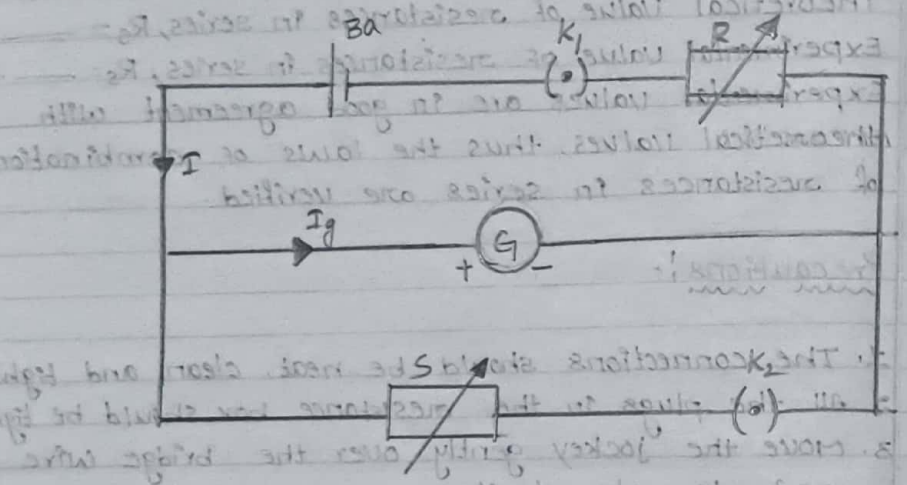
1. The connections should be neat, clean and tight.
2. All the plugs in the resistance box should be tight.
3. Move the jockey gently over the bridge wire and do not rub it.
4. The plug in key K should be inserted only when the observation to be taken.

Source of error:-

1. connections may be loose.
2. The wire may not have uniform thickness.

Teacher's Signature: .....

Circuit Diagram:-



Where:

Ba – Battery

G – Galvanometer

R & S – Resistance boxes

K1 & K2 – plug keys

FIGURE * OF * MERIT

Expt. No.: 4 Date : ..... Page No.: 11

Aim:-

To determine resistance of a galvanometer by half deflection method and to find its figure of merit.

Apparatus:-

* Galvanometer
* Battery
* Two resistance boxes
* One way key (2 numbers)
* Connecting wires.

Theory:-

A galvanometer is a device used to detect small current in a circuit. It has a coil pivoted in a radial magnetic field. When electric current is passed through the coil, it gets deflected. Its deflection is noted by attaching a pointer to the coil. The deflection is proportional to current passed. A galvanometer has a moderate resistance and has a small method. The circuit is shown for this method. key K is inserted and deflection θ is determined with a suitable value of R. If E is the emf of cell and I be the current in circuit, the galvanometer resistance.

$$G = \frac{RS}{R-S}$$

Teacher's Signature: .....

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Observation:-

S.NO	Resistance Ω	Deflection galvanometer θ (divisions)	$\frac{\theta}{2}$ (div)	Shunt S Ω	Galvanometer Resistance G Ω	Figure of merit $K=\frac{E}{(R+G)\theta}$

Mean (K) = _____ A/division

Expt. No.: .....

Date : .....

Page No.: 12

Figure of merit of galvanometer. It is defined as current required for producing deflection of 1 division. It is measured in ampere/div. When a high resistance R is taken out from resistance box, a current I flow in the circuit and it Produces a deflection θ.

$$I = \frac{E}{R+G} = K\theta$$

Therfore

$$K = \frac{E}{(R+G)\theta}$$

By varying R and noting corresponding value of θ, we can find a set of values of figure of merit. The mean of these values gives the figure of merit.

Procedure:-

a) Make the connections as shown.
b) See that the plugs of resistance box are tight
c) Introduce a resistance of 5000 From the resistance box R and then insert the KI only
d) Adjust the value of R so that the deflection in the galvanometer is maximum (say 30 div).
e) Note the deflection. Let it be θ i.e, θ = 30 div.
f) Now, insert the key K2. Without changing the value of R, adjust the value of K, such that deflection in galvanometer reduces exactly to half the value obtained i.e, θ/2.

Teacher's Signature: .....

Scanned with OKEN Scanner

Expt. No.: ..... Date : .....
Page No.: 13

g) Note the value of R.
h) Repeat the above steps for various values of R.

Result:-

1) The resistance of galvanometer is found to be
2) Figure of merit of galvanometer is _____ A/div

Precautions:-

a) All the connections should be neat, clean & tight
b) Ensure that the plugs of resistance box are tight
c) Initially a high resistance from the resistance box should be introduced or else a small resistance can damage the galvanometer.

Sources of error:-

1. The emf of the battery may change during the experiment.
2. Calibration of resistance in resistance boxes may not be correct.
3. Plugs in resistance boxes may not be tight and may have contact resistance

Teacher's Signature: .....

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Diagram :-

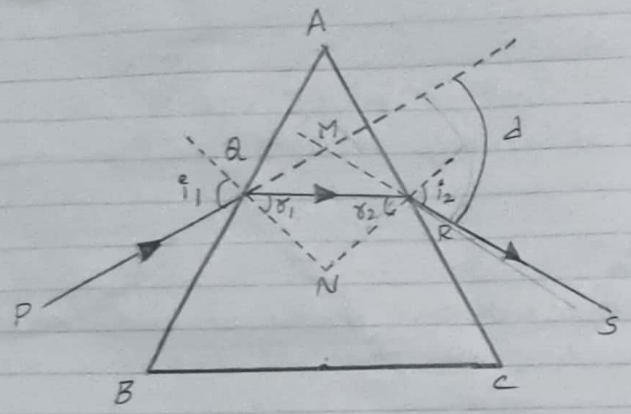
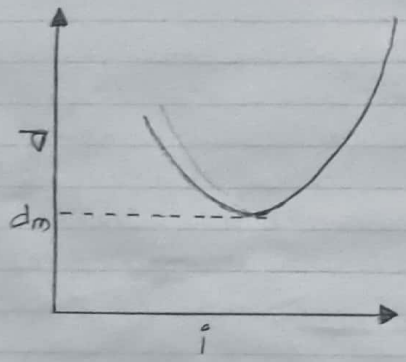


Figure – 1

Nature of Graph :-



ANGLE OF MINIMUM DEVIATION

Expt. No.: 5 Date : .....

Page No.: 14

Aim :-

To determine the angle of minimum deviation for a given glass prism by plotting graph between angle of incidence and the angle of deviation.

Apparatus :-

* Drawing Board
* Prism
* Drawing Pins
* White Paper

Principle :-

A ray of light incident on one of the refracting surface of prism of angle A and refractive index n, it will deviate while passing through Prism as shown in Figure-1. The deviation is given by

$$d = (i_1 + i_2) - A$$

Where i_1 and i_2 are angle of incident and angle of emergence.

As the angle of incidence is gradually increased, the angle of deviation first decreases reaches a minimum value (d_m) and then increases as shown in graph. When the angle of incidence is equal to the angle of emergence, the deviation is minimum.

Teacher's Signature: .....

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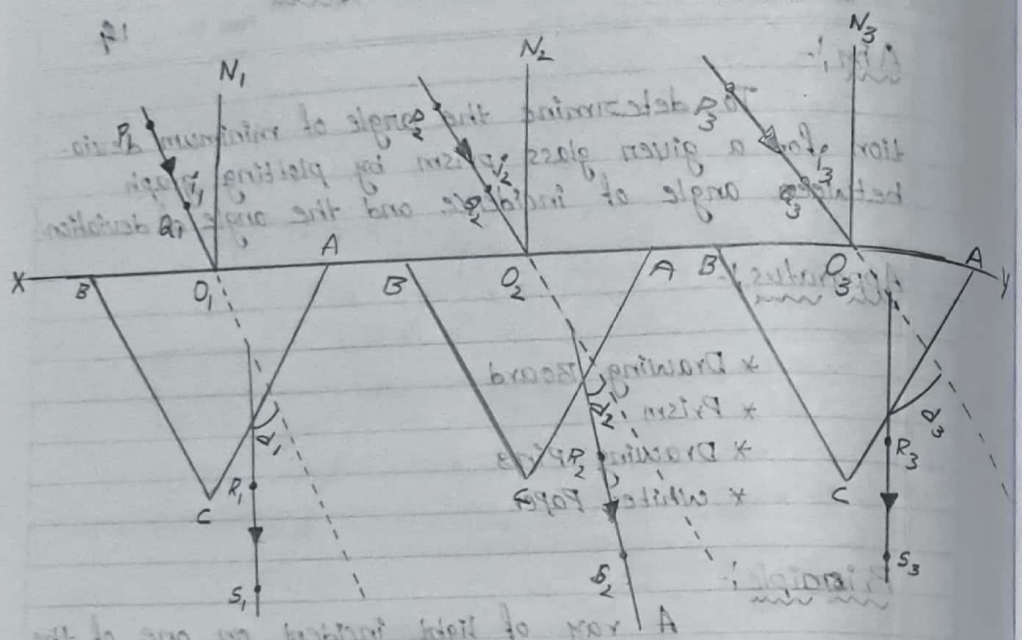


Figure - 2

Where:-

ABC - prism,

$A-(i_1+i_1)=b$

$P_1Q_1R_1S_1$, $P_2Q_2R_2S_2$ ---- Pins;

i_1, i_2 & i_3 - different angle of incidence,

d_1 d_2 & d_3 - angle of deviation

Expt. No.: .....

Date : .....

Page No.: 15

Procedure:-

1) A straight line xy is drawn on the sheet of White Paper fixed on the drawing board.

2. The Points O_1, O_2, O_3 ---- are marked on line xy.

3. A normal N_1O_1 is drawn to xy at O_1 and straight line D_1O_1 is drawn to represent ray of incidence which makes an angle 45^0 with N_1O_1.

4. The Prism ABC is placed on the paper as shown in the Figure and its boundary is drawn.

5. Two pins P_1 and Q_1 are vertically fixed about 5cm apart on the ray of incidence D_1O_1.

6. While seeing the images of the pins P_1 and Q_1 through the other refracting surface BC of the prism, two more pins R_1 and R_2 are fixed on the side of AC so that R_1S_1 and the images of the pins P_1 and Q_1 will be collinear.

7. Pins are removed and their pricks ar marked. A straight line is drawn through the pin Pricks of R_1 and S_1 to obtain the ray of emergence.

Teacher's Signature: .....

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Observation :-

S.NO	Angle of incidence i	Angle of deviation d
1.		
2.		
3.		
4.		
5.		
6.		

$$\mu = \frac{\sin\left(\frac{A+Dm}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Calculation :-

From graph

Angle of minimum deviation dm = _____

Expt. No.: .....

Date : .....

Page No.: 10

8. The ray of incidence and emergence are extended as shown in the figure-2 to find the angle of deviation (d)

9. The experiment is repeated for different value of angle of incidence and to find the corresponding angle of deviation.

10. A graph of angle of incidence (i) versus angle of deviation (d) is drawn and angle of minimum deviation dm is found from the graph.

Result :-

The angle of minimum deviation, d_m = _____

Precautions :-

1. The angle of incidence should lie between 35°-60°.
2. The pins should be fixed vertically.
3. The distance between the two pins should not be less than 10 mm
4. Arrow heads should be marked to represent incident and emergent rays
5. The same angle of prism should be used for all observations.

Teacher's Signature: .....

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Expt. No.: .....

Date : .....

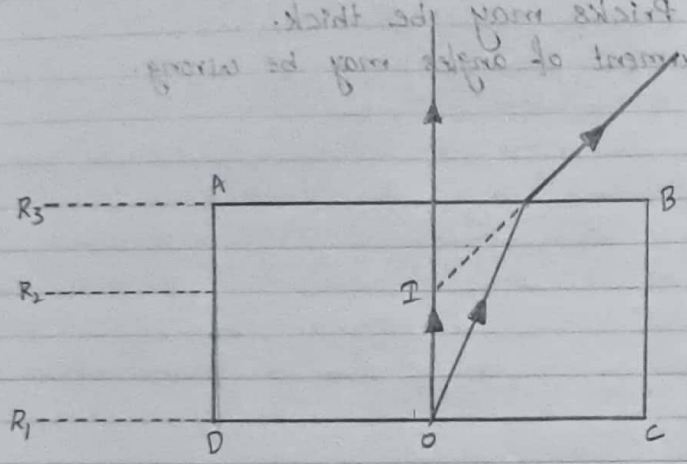
Page No.: 17

Source of error :-

1. Pin Pricks may be thick.
2. Measurment of angles may be wrong.

Teacher's Signature: .....

Diagram:-



Where:

ABCD - Rectangular glass slab

O - Object

I - Image

R_3 - position of the particles on glass slab.

R_2 - Position of the image [object seen through glass slab]

R_1 - Position of the object.

REFRACTIVE INDEX OF GLASS

Expt. No.: 6. Date : .....

Page No.: 18

Aim:-

To determine the refractive index of glass using travelling microscope.

Apparatus:-

* Travelling microscope
* A glass slab
* chalk dust
* Paper

Principle:-

If a glass slab is placed in air on horizontal surface and its bottom surface is viewed from top, it appears to be elevated due to the refraction. The distance between the apparent bottom and the top surface of the slab gives the apparent thickness of the slab. In case of normal observation, the refractive index of the glass with respect to air is the ratio of real thickness of glass slab to the apparent thickness.

Formula:-

1. Total reading $TR = MSR + (CVD \times LC)$. cm

2. Refractive index of glass slab $n_g = \frac{R_3 - R_1}{R_3 - R_2}$

Teacher's Signature: .....

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Observations:-

The value of the LMSD of travelling microscope (S) = _____ cm

Total number of USD (N) = _____

Least count (LC) = $\frac{\text{The value of LMSD}}{\text{Total no of USD}}$ = _____ = _____ cm

S. No	Travelling Microscope Reading for								
	Mark made on paper			Mark made on through glass slab			Particles on the top of glass surface		
	MSR in cm	CVD in div	TR in cm (R_1)	MSR in cm	CVD in cm	TR in cm (R_2)	MSR in cm	CVD in div.	TR in cm (R_3)

Calculation:-

Refractive index of the slab = $\frac{\text{Real thickness of the slab}}{\text{Apparent thickness of the slab}}$

$$n_g = \frac{R_3 - R_1}{R_3 - R_2} = \frac{b}{c}$$

Expt. No.: .....

Date : .....

Page No.: 19

Where:

MSR – Main scale Reading

CVD – coincide vernier scale reading

LC – Least count

R_1 – Travelling microscope reading when focused on object (ink mark) through air.

R_2 – Travelling microscope reading when focused on image when glass slab is kept on object

R_3 – Travelling microscope reading when focused on the Particles on the surface of glass slab

Procedure:-

1. The least count of travelling microscope is calculated.

2. A sheet of white Paper with ink mark is placed on the base of travelling microscope

3. The microscope is adjusted to focus on the ink mark on the paper & readings are noted in

Teacher's Signature: .....

Scanned with OKEN Scanner

Expt. No.: .....

Date : .....

Page No.: 20

Vertical scale and total reading (R_1) is found using relevant formula.

4. Now glass slab is placed on the paper and the microscope is focused on the ink mark through glass slab and corresponding reading R_2 is found.

5. Some chalk dust is sprinkled on the upper surface of the glass slab the microscope is focused on chalk dust & corresponding reading R_3 is found.

6. Refractive index of glass slab is calculated using the formula

$$n_g = \frac{R_3 - R_1}{R_3 - R_2}$$

7. The experiment is repeated for one more trail to raise the position of glass slab and hence calculate the refractive index. Then mean refractive index is found.

Result:-

The refractive index of glass is (n_g) = _____

Precautions

1. In microscope, the parallax should be properly removed.
2. The microscope should be moved in upper direction only to avoid back lash error.

Source of error

1. The microscope scale may not be properly calibrated.

Teacher's Signature: .....

Diagram:

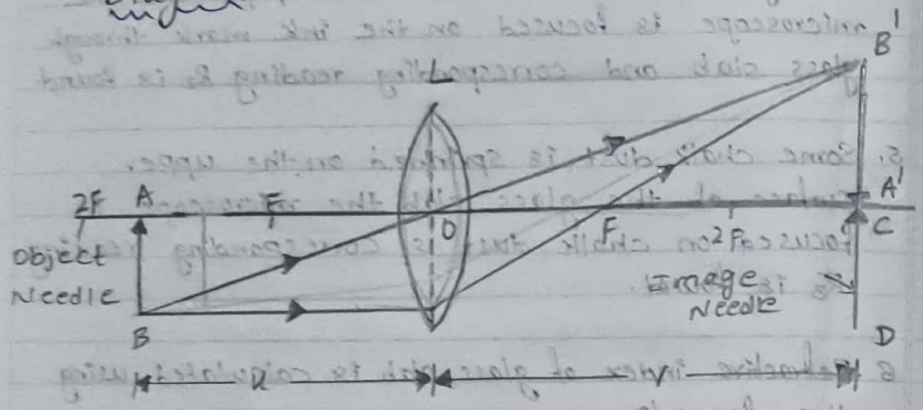
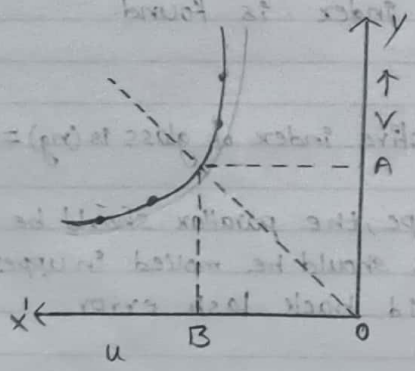


Fig. Focal length of convex lens

Nature of Graph;-



Focal Length of Convex Lens

Date : .....

Expt. No.: .....

Page No.: 21

Aim;- Determination of the focal length of convex lens by plotting a graph of u versus v.

Apparatus;- optical bench, two sharp edged pin, convex lens of lens than 20cm focal length, three upright with clambs & meter scale.

principle;- 1. The object distance must be greater than focal length of the convex lens to get real and inverted image.

2. When object distance is equal to twice the focal length of the convex lens, then the image distance is equal to twice the focal length.

Formula;-

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{u-v}{uv}$$

$$f = \frac{uv}{u-v}$$

Where; f = focal length of convex lens

u = distance of object needle from optic centre of lens

v = distance of image needle from optic centre of the lens

Procedure;-

1. The given convex lens is moved forward & backward to obtain the sharp image of the distance tree on the white sheet of paper kept vertically behind the lens. The distance b/w the lens & white sheet of paper

Teacher's Signature: .....

Observation

S.NO	Object distance (U in cm)	Image distance (V in cm)
1.		
2.		
3.		
4.		
5.		
6.		

is measured. This is the approximate Focal length of lens

2) The upright mounted with convex lens, object pin P_1 and image pin P_2 are placed on one side of the optical bench as shown in the Fig.

3) The heights of the object pin P_1 & image pin P_2 are adjusted such that their tips lie on the principle axis of the lens.

4) The position of the object pin P_1 is adjusted such that it will be at a distance which is greater than of the lens

5) Experiment is repeated at different values of u & readings are tabulated.

6) A graph is plotted blw U and V, taking U on x-axis & V- on y-axis. The angular bisection OZ is drawn and OA & OB are measured

7) Focal length of the convex lens is calculated using the formula

$$f = \frac{OA + OB}{4}$$

Result

The Focal length of the convex lens, f = _____ cm

Precautions:-

1. Parallax should be removed From tip to tip by keeping eye at a distance at least 30cm away from needle

Teacher's Signature: .....

Expt. No.: .....

Date : .....

Page No.: 23

2) The object needle should be placed at such a distance that only real, inverted image of it is formed.

3) Index correction for u and v should be applied.

4) Tipb of the object & image needles should lie at the bame height ab the centre of the lens

Source of error ;-

1. The uprights may not be the vertical

2- Parallax removal may not be perfect

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Diagram:-

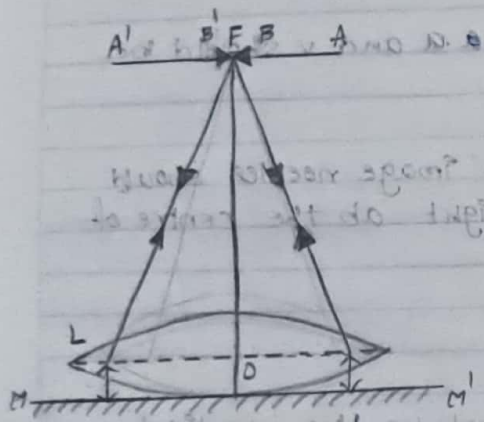


Figure-1

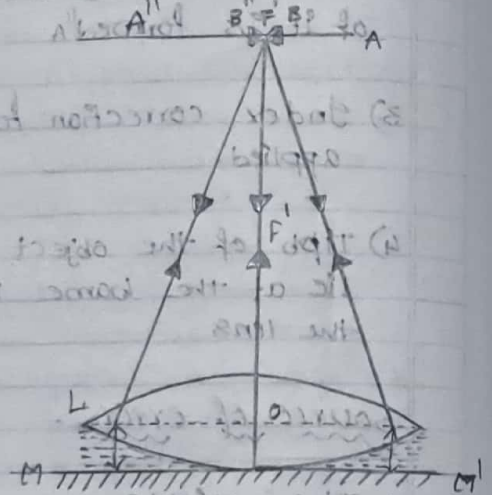


Figure-2

Where:-

L - Convex lens

MM' - plane mirror

AB - object pin

A'B' - image by convex lens

A''B'' - image by combination of water lens and convex lens

O - optic centre

Expt. No.: 8 Refractive Index of water By Convex Lens Date : Page No.: 24

Aim:- To determine the refractive index of water using convex lens & plane mirror

Apparatus:- plane mirror, convex lens, laboratory stand with rigid base and clamp arrangement, water, a pin, a plumb line, spherometer and a meter scale.

Principle:- The reciprocal of the equivalent focal length of the combination of the lens is the equal to the sum of the reciprocal of focal length of individual lenses

In figure-1, OF = f is the focal length of the convex lens. In fig-2, OF = f' is the equivalent focal length of combination of convex lens & the plano concave lens of water. If f_w is the focal length of plano-concave lens of water.

$$\frac{1}{f'} = \frac{1}{f} + \frac{1}{f_w} \Rightarrow \frac{1}{f_w} = \frac{1}{f'} - \frac{1}{f}$$

$$\therefore f_w = \frac{ff'}{f-f'}$$

From lens maker formula the focal length of plano-concave water lens:

$$\frac{1}{f_w} = (n_w - 1)\frac{1}{R}$$

$$\therefore n_w = 1 + \frac{R}{|f_w|}$$

Formula:-

1. Focal length of plano-concave water lens $f_w = \frac{ff'}{f-f'}$
2. Refractive index of water. $n_w = 1 + \frac{R}{f_w}$

where: f - Focal length of convex lens.

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Observation :-

1. The radius of curvature of the given convex lens (R) = _____ cm

To measure the Focal length of water lens :-

S. NO	Position of pin w.r.t the optical centre O						$f_w = \frac{ff'}{f-f'}$ in cm
	Distance of the pin without water from		$f = \frac{d_1+d_2}{2}$ in cm	Distance of the pin with water from		$f' = \frac{d_1'+d_2'}{2}$ in cm	
	Upper surface of the lens (d_1 cm)	plane mirror (d_2 cm)		Upper surface of the lens (d_1' cm)	plane mirror (d_1' cm)		

Mean f_w = _____ cm

Calculation

* Focal length of the plano-concave lens of water f_w = _____

* Refractive index of water $n_w = 1 + \frac{R}{f_w}$

Expt. No.: .....

Date : _____

Page No.: 25

Procedure :-

1. A plane mirror is placed on the base of laboratory stand keeping its reflecting surface upward.
2. A convex lens is placed on the plane mirror.
3. A sharp edged bright pin AB is placed horizontally just above the optic centre (O) of the convex lens and clamped
4. The position of the pin AB is adjusted such that it coincide with its image A'B' without parallax.
5. The vertical distance blw the optic centre (O) of convex lens and pin is measured using plumb line. It's equal to focal length f of convex lens.
6. The position of the pin AB is raised such that it coincide with its image A''B'' without parallax
7. The Focal length of water lens is calculated using the relevant formula
8. The radius of curvature of the spherical surface of the convex lens in contact with water is found by using spherometer.
9. Refractive index of water is calculated using formula

Result :- The Focal length of the plano-concave lens of water f_w = _____

Precautions :- 1. The parallax should not be there
2. Meter scale should be straight while taking reading
3. Air bubbles should not be there in water.

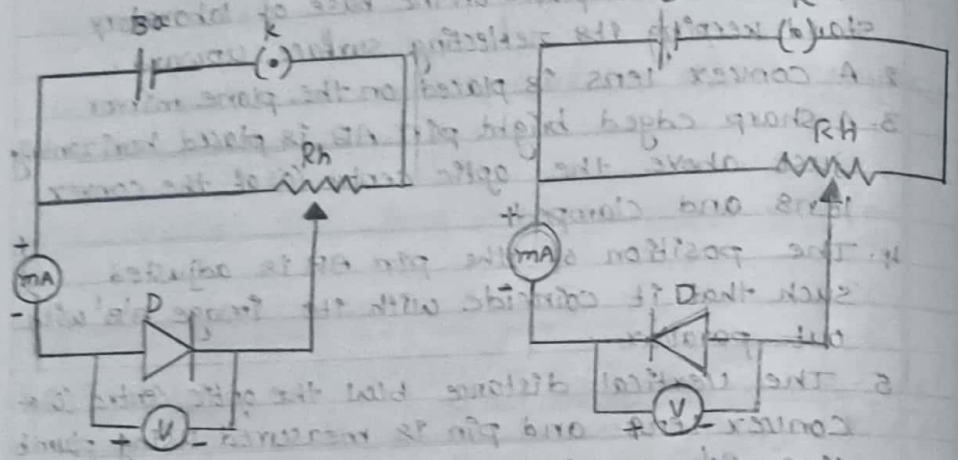
Source of error

1. Formation Air bubbles in water
2. Impurities in water.

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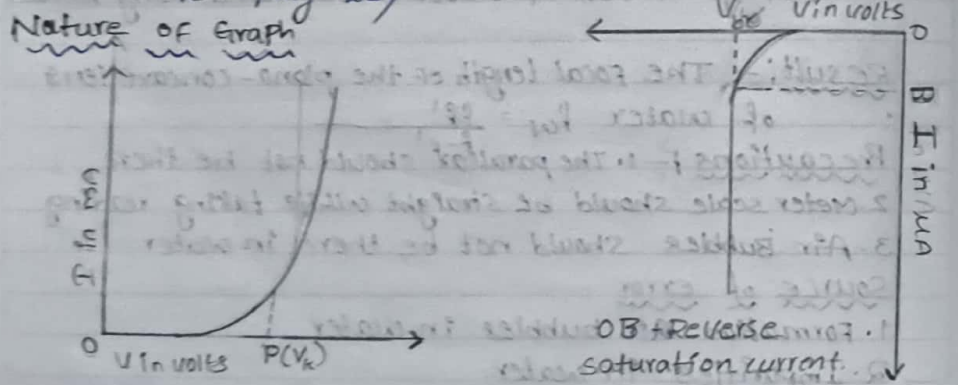
Circuit Diagram



Forward Bias Reverse Bias

Where: Ba – Battery
V – Voltmeter
μA – Micro ammeter
mA – Milli ammeter
Rh – Rheostat
D – Diode
K – plug key

Nature of Graph



Forward Bias Reverse Bias

SEMI-CONDUCTOR DIODE

Expt. No.: 9 Date : Page No.: 26

Aim:- To draw the I-V characteristic curve of a P-n junction in forward bias and reverse bias.

Apparatus:- A P-n junction diode, milli ammeter, microammeter, voltmeter, rheostat & battery.

Principle:- When P-side of the semi-conductor diode is connected to the positive terminal of the battery then P-n junction is said to be forward biased. The junction offers low resistance and initially a negligibly small current flows through circuit till the applied voltage crosses a certain value the diode current increases exponentially.

When P-side of the semi-conductor diode is connected to the negative terminal of the battery, then P-n junction is said to be reverse biased. As the applied voltage is increased in the reverse biased condition, starting from zero, the current increases, but soon become constant. It is called reverse saturation current.

Procedure

(a) For Forward bias:-

1) The connections are made as shown in the circuit
2) Using rheostat, the voltage is adjusted for a value V. the voltage V and corresponding current I are noted.

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Observations:-

Forward Bias characteristics:-

S.NO	Voltage (V) in volts	current I in mA

Reverse Bias characteristics:-

S.NO	Voltage (V) in volts	current I in mA

Expt. No.: .....

Date : .....

Page No.: 27

3. The voltage is increased in small steps. value of V and I are noted in each case and readings are tabulated.
4. A graph is plotted for the current I versus voltage V.
5. cut in voltage is located in the graph.

(b) For reverse bias:-

1. The connections are made as shown in the circuit
2. Using rheostat, the voltage is adjusted for a value V. the voltage V and corresponding current I are noted.
3. The voltage is increased in small steps. value of V and I are noted in each case & reading are tabulated.
4. A graph is plotted for the current I versus voltage V.
5. Reverse saturation current is found.

Result:-

I-V characteristics of p-n junction diode are drawn

From graph: cut in voltage = Op = _____ V

Precautions:-

1. The connections should be neat, clean & tight.
2. Forward-bias voltage beyond breakdown should not be applied
3. Reverse-bias voltage beyond breakdown should not be applied.

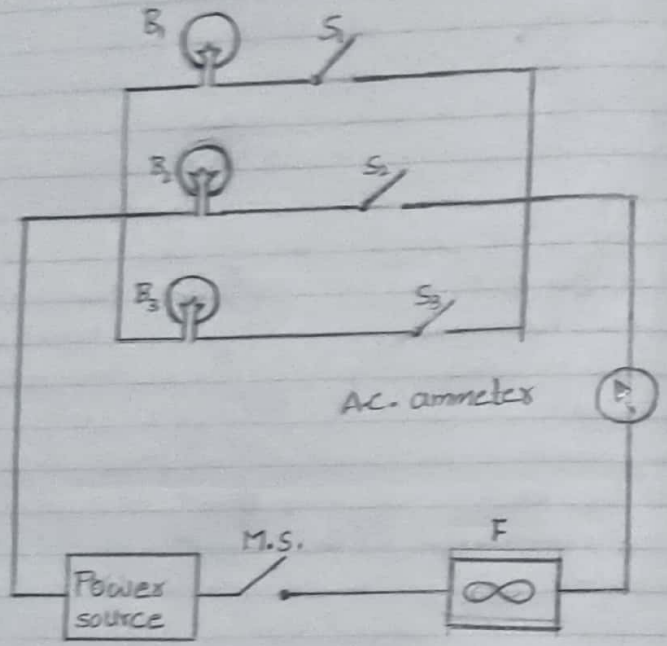
Source of error:-

The junction diode supplied may not be faulty.

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Diagram:-



(a) circuit diagram

Expt. No.: _____ **Activity-1** Date : _____ Page No.: 28

Aim:-

To assemble a household circuit comprising three bulbs, three (on/off) switches, a fuse and a Power source.

Apparatus and material:-

Apparatus:-

No apparatus required in assembling in circuit.

Material:-

* Three bulbs (6V, 1W) each
* Fuse of 0.6A
* Main switch a power supply (battery eliminator)
* Three (on/off) switches Flexible connecting wire with red and black plastic covering
* A Fuse wire

Supplementary:-

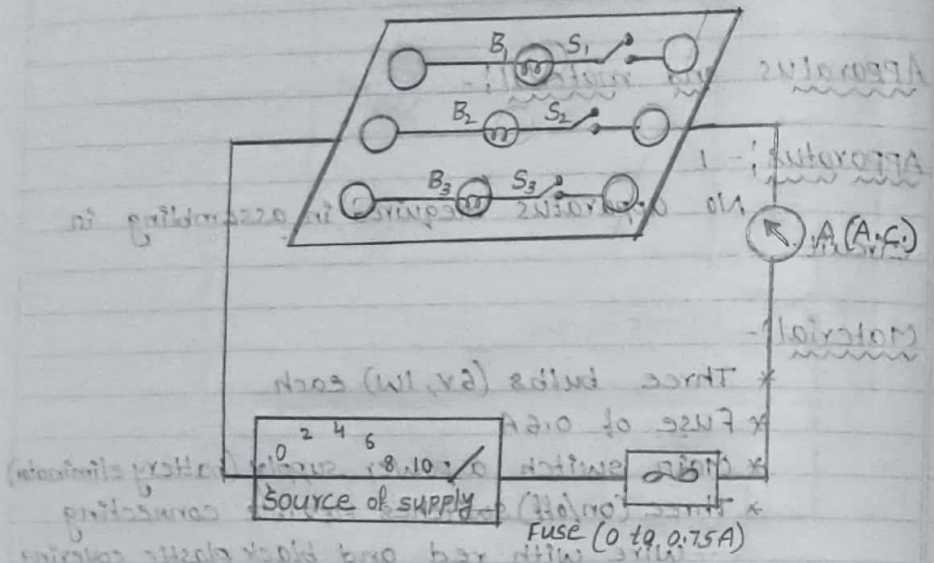
Main electric board with a two-pin socket and main switch.

Theory:-

Electricity supplied to us for domestic purpose is 220V A.C. and 50 Hz. The household circuit, all appliances are connected in 'parallel' with mains. The switches are connected in series with each appliances in live wire. 5A switches are required for normal appliances like,

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(b) Actual layout

Expt. No.: .....

Date : .....

Page No.: 29

bulbs, Fluorescent tubes fans etc. 15A sockets and switches are required for heavy load appliances like, refrigerator, air conditioner, geyser, hot Plates etc. All appliances must have three wires called live, neutral and the earth. Total Power consumption 'P' at a time.

$P = P_1 + P_2 + P_3 + \cdots$

where P_1, P_2, P_3 are the powers drawn by appliances.

To protect the appliances from damage when unduly high currents are drawn fuse of little higher rating, 10 to 20% higher than the current normally drawn by all appliances. For further safety, a suitable value MAINS FUSE like rating 32A is connected in series with supply source.

Procedure :-

1. Connect the bulbs B_1, B_2 and B_3 in series with switches S_1, S_2 and S_3 respectively and connect each set of B-S in parallel with each other.

2. Connect main supply to a step-down transformer (battery eliminator) to get.

3. Connect the mains fluse M.S. in series with the power supply (battery eliminator).

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Expt. No.: .....

Date : .....

Page No.: 30

4. Connect an A.C. ammeter in series with the B-S set.

5. Connect one end of power supply to one end of B-S set.

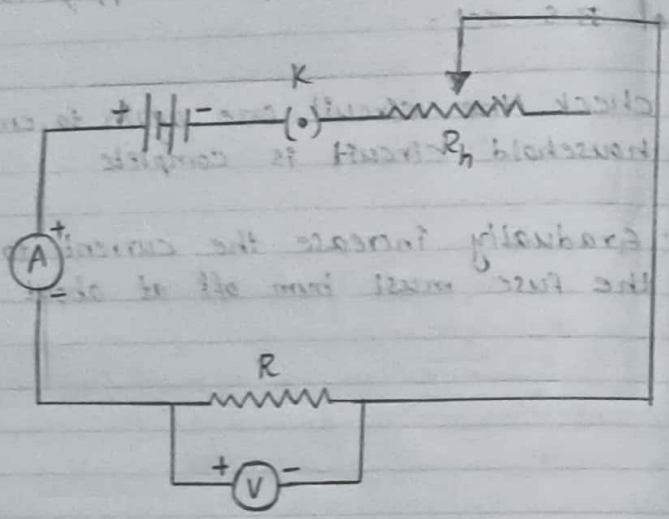
6. Check the circuit one again to ensure that household circuit is complete.

7. Gradually increase the current to 0.75 A, the fuse must bum off at about 0.6 A.

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Diagram :-



Expt. No.: .....

ACTIVITY-2

Date : .....

Page No.: 31

Aim :-

To assemble the components of a given electrical circuit.

Apparatus / material required :-

* A voltmeter
* An ammeter of appropriate range
* A Battery
* A rheostat
* one way key
* An unknown resistance or resistance coil
* connecting wires
* a piece of sand paper

Procedure :-

1. Connect the components (Resistors, inductors etc.) in series with each other as shown in diagram and then in series with the battery.

2. Connect the ammeter in series with the circuit, to measure the current.

3. Connect the voltmeter in parallel to the circuit, to measure the potential difference.

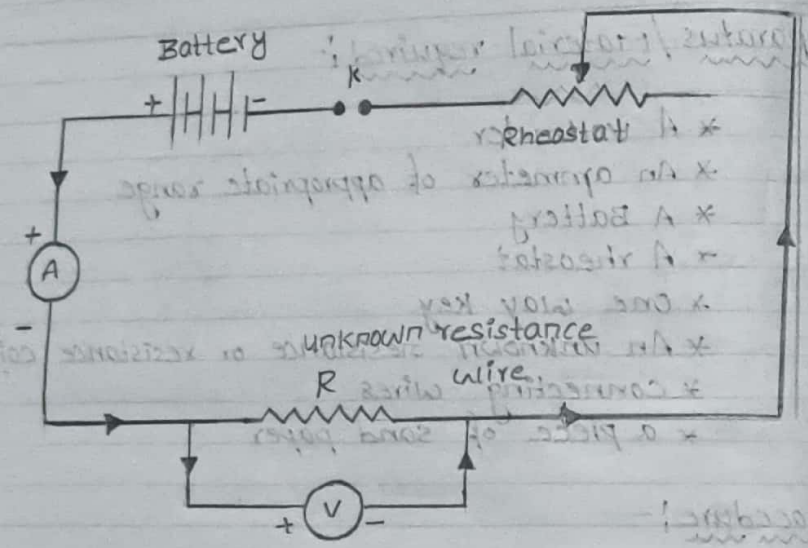
4. Connect the switch in series with the battery

5. Assembly of the electrical componets in electric circuit is complete.

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Circuit Diagram:



Activity-3

Expt. No.: 3

Date :

Page No.: 32

Aim:- To drawn the diagram of a given open circuit comprising at least a battery, resistor/ rheostat, key, ammeter and voltmeter. Mark the components that are not connected in proper order and correct the circuit and also the circuit diagram

Apparatus:-

* Battery eliminator (0 to 6V)
* Rheostat
* Resistance box (0 to 100Ω)
* Key, D.C. ammeter (0-3)A
* D.C. voltmeter (0-3)V

Theory:-

An open circuit is the combination of a primary components of electric circuit in a such a manner that on closing the circuit no current is drawn from the battery

Procedure:-

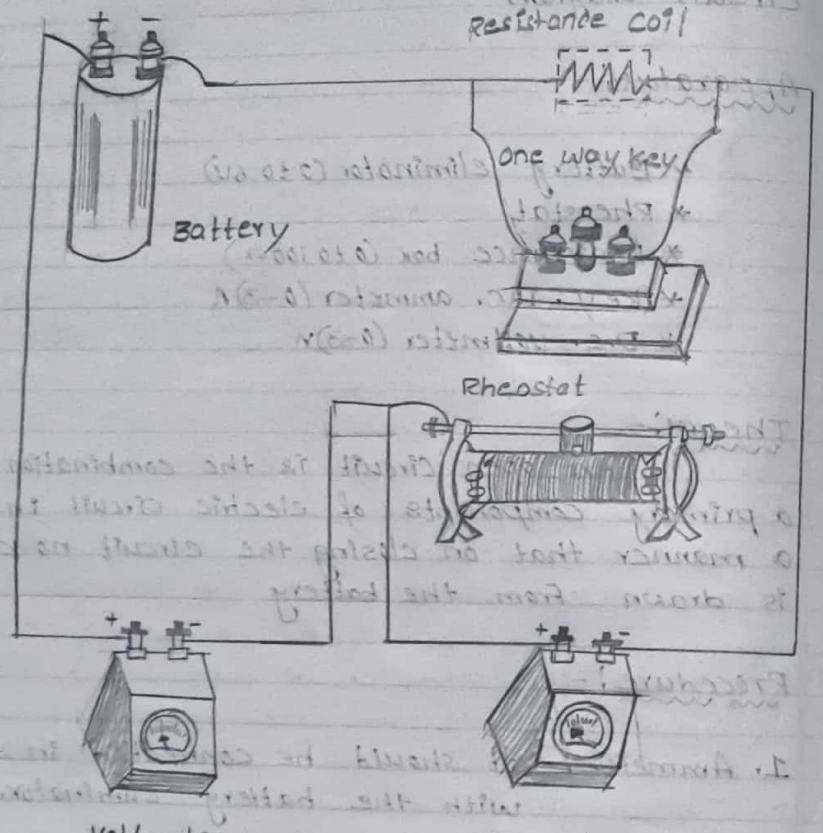
1. Ammeter:- It should be connected in series, with the battery eliminator.

2. Voltmeter:- It should be connected in parallel to the resistor

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Diagram :-



Open circuit diagram (components not connected in proper order)

Expt. No.: .....

Date : .....

Page No.: 35

3. Rheostat :- It should be connected in series (in place of resistance coil) with the battery eliminator.

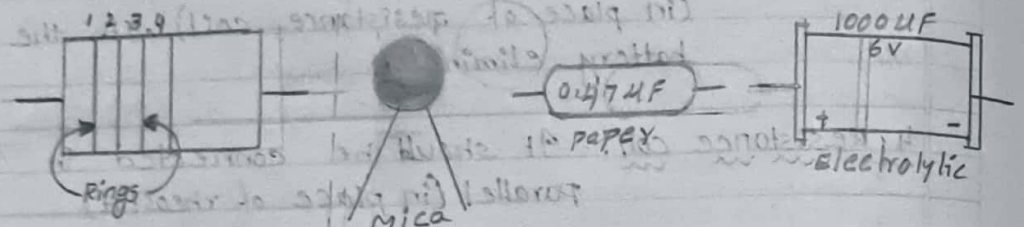
4. Resistance coil :- It should be connected in parallel (in place of rheostat)

5. One way key :- It should be connected in series to the battery eliminator.

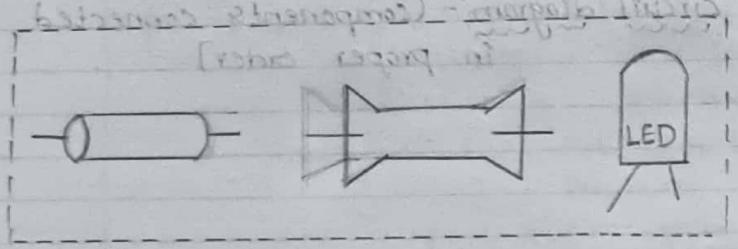
6. Correct circuit diagram :- (Components connected in proper order)

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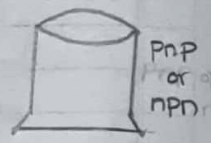
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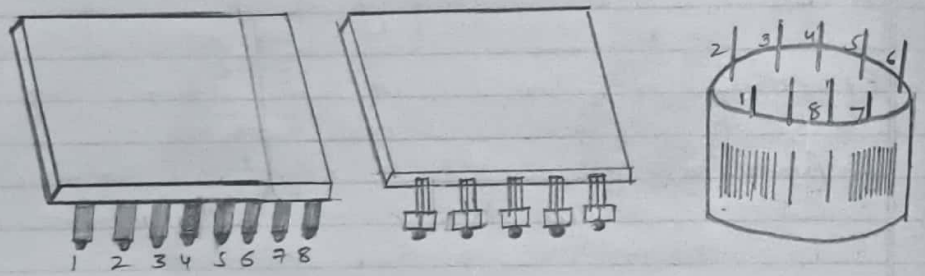
(a) carbon resistor (b) capacitors



(c) Diode



(d) Transistor



(e) Integrated circuits

Fig. Some of the commonly available integrated circuits (ICs)

Expt. No.: ACTIVITY-4 Date :

Page No.: 34

Aim :-

To identify a diode, an LED, a transistor, an IC, a resistor and a capacitor from a mixed collection of such items.

Apparatus and material :-

Apparatus :- Multimeter

Material :- Above mixed collection of items.

Theory :-

For identification, appearance and working of each item will have to be considered.

1. **Diode :-** A diode is a two terminal device. It conducts when forward biased and does not conduct when reverse biased. It does not emit light while conducting. Hence, it does not glow.

2. **LED :-** A LED (Light emitting diode) is also a two terminal device. It also conducts when forward biased and does not conduct when reverse biased. It emits light while conducting. Hence, it glow.

3. **Transistor :-** Transistor is a three terminal device. The terminals represent emmitter (E), base (B) & collector (C).

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Observation:-

No. of Obs.	Number of legs	Name of device	No. of Obs	Possible current flow	Name of device
1.	More than 3	IC	4.	Unidirectional emit no light	Diode
2.	Three	Transistor	5.	Unidirectional emit light	LED
3.	Two	Capacitor, Diode, LED or resistor	6.	Both direction (steady)	Resistor
			7.	Intial high but decay to zero	capacitor

Expt. No.: .....

Date : .....

Page No.: 35

4. IC :- An IC (integrated circuit) is a multi-terminal device in form of a chip. [see figure (UM 3482 IC Tone Generator)]

5. Resistor : A Resistor is two terminal device. It conducts when either forward biased or reverse biased. (Infact there is no forward or reverse bias for a resistor). It conducts when operated with A.C. voltage.

6. capacitor : A capacitor is also a two terminal device. It does not conduct when either forward biased or reverse biased. when a capacitor is connected to a D.C. source, then multimeter shows full scale current intially but it decay to zero quickly. It is because that intially a capacitor drawn a charge.

The components to be identified are shown in figure

Procedure :-

1. If the item has four or more terminals and has form of a chip, it is an IC (integrated circuit).

2. If the item has three terminals, it is a transistor

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Expt. No.: .....

Date : .....

Page No.: 36

3. If the item has 2 terminals, it may be diode, a LED, a resistor or a capacitor.
To differentiate proceed as ahead.

4. Put the selector on resistance R of multimeter for cheking the continuity. The probe metal ends are inserted in terminal marked on the multimeter as common and P (or +ve).

If such that the black one is in common and red probe is in P (or +ve). On touching the 2 ends of the device to the 2 other metal ends of Probes.

1. If Pointer moves when voltage is applied in one way and does not move when reversed and there is no light emission, the item is a diode.

2. If Pointer moves when voltage is applied in one way and does not move when reversed and there is light emission, the item is a LED.

3. If Pointer moves when voltage is applied in one way and also when reversed, the item is a resistor.

4. If Pointer does not move when voltage is applied in one way and also when reversed, then item is a capacitor.

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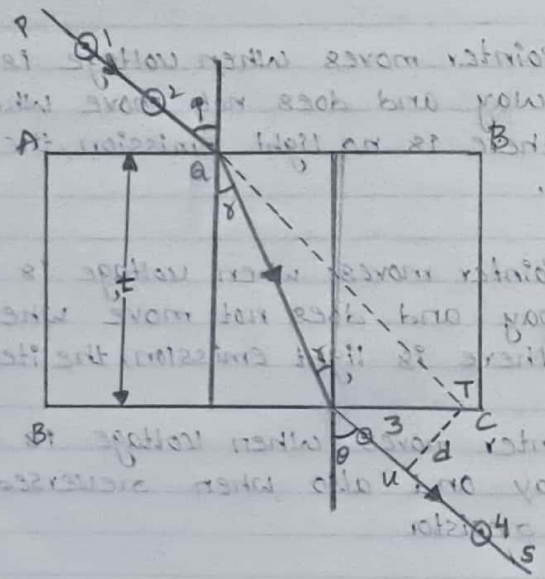
Observation :-

Angle of incidence, i = _____

Angle of refraction, r = _____

Thickness of glass slab, t = AD = _____ cm

Lateral displacement, d = TU = _____ cm



ACTIVITY-5

Expt. No.: .....

Date : .....

Page No.: 37

Aim :-

To observe refraction & lateral derivation of a beam of light incident on a glass slab.

Apparatus and material Required :-

* Glass slab (size 10×6 cm)
* Drawing board
* Drawing Pins
* Plane sheet of paper
* Pins
* Common scale
* Protractor
* Pencil
* Rubber
* Cello tape
* sharply pointed pencil
* Half meter scale

Procedure :-

1. Fix a white paper sheet by drawing pins and drawing.

2. Take a glass slab and put it symmetrically in the middle of the paper and mark its boundary ABCD

3. Draw a normal at point Q on face AB and draw a line PQ making an angle i with the

Teacher's Signature: .....

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Expt. No.: .....

Date : .....

Page No.: 38

normal. PQ will respect an incident ray.

4. Fix two points 1 and 2 on the line PQ at distance of 10cm or more between themselves.

5. See images of these pins through face DC and fix two more pins at points 3 and 4 (10 cm or more apart) such that these two pins cover the images of first two pins, all being along a straight line.

6. Remove the glass slab. Draw straight line RS through points 3 and 4 to present emergent ray. Join QR to represent refracted ray.

7. Draw normal at point R on face DC and measure angle e. It comes to be equal to i.

8. Produce PQ forward to cut DC at N. Draw RK Perpendicular to PN. RK measures lateral displacement at d.

Calculations:-

Theoretical value of displacement $d = t \sec r \sin(i-r)$ _____ cm

Different between measured and theoritical value of lateral displacement = _____ cm

Teacher's Signature: .....

Expt. No.: .....

Date : .....

Page No.: 29

Conclusion/Result:-

1. The ray of light emitting out from a glass slab is parallel to the direction of incident ray, but is laterally deviated.

2. The angle of incident is equal to the angle of emergence, i.e. $i = e$.

3. The lateral deviation of emergent ray w.r.t. incident ray is directly proportional to the thickness of glass slab.

Precautions:-

1. The upper surface of drawing board should be soft, so that the pins can be penetrated easily.

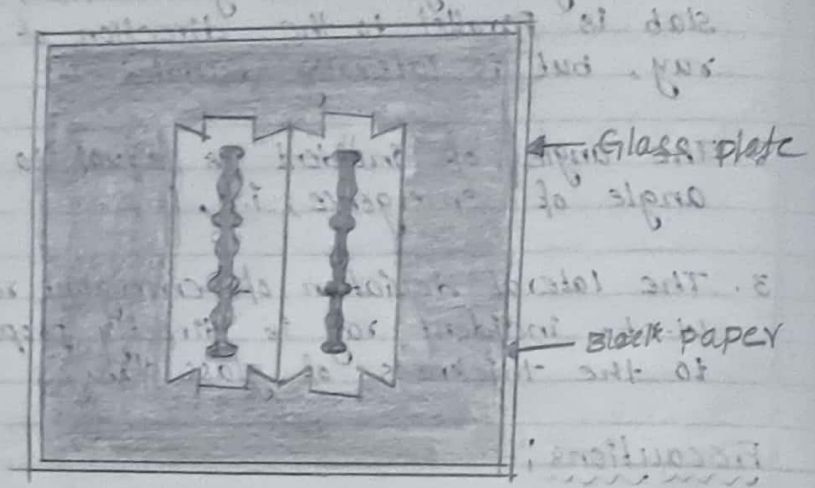
2. The all pins must be vertically fixed on the board.

3. A sharp pencil should be used to draw boundary of the glass slab.

4. The glass slab should be of uniform thickness

5. To locate the incident ray and emergent ray with greater actually, the distance between pins should be 6 cm to 8 cm.

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A thin slit made by using two razor blades, black paper and a glass plate

ACTIVITY-6

Expt. No.: .....

Date : .....

Page No.: 40

Aim:-

To observe diffraction of light due to a thin slit.

Apparatus:-

* Two razor blades
* Adhesive tapes
* Screen a source of monochromatic light (laser pencil) black paper
* Glass plate

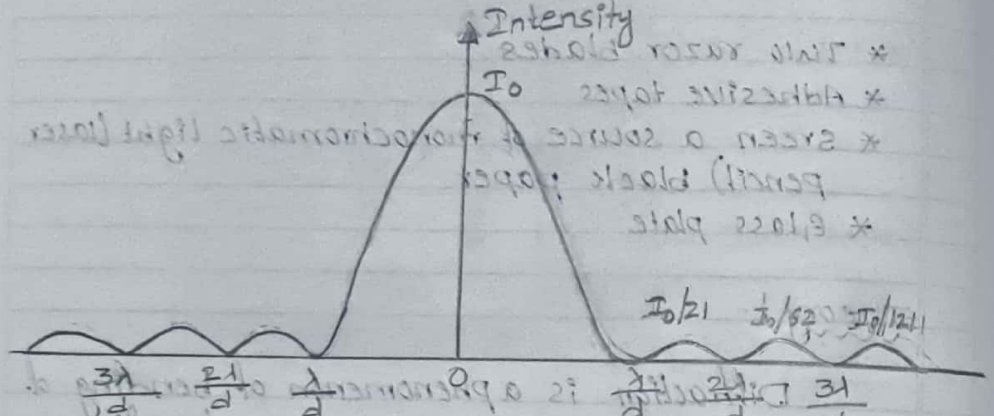
Theory:-

Diffraction is a phenomenon of bending of light around the corners or edges of a fine opening or aperture. Diffraction takes place when order of wavelength is comparable or small to the size of slits or aperture. The diffraction effect is more pronounced if the size of the aperture or the obstacle is of the order of wavelength of the waves.

The diffraction pattern arises due to interference of light waves from different symmetrical point of the same wavefront. The diffraction pattern due to a single slit consists of a central bright band having alternate dark and weak bright bands of decreasing intensity on both sides.

Teacher's Signature: .....

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Expt. No.: .....

Date : .....

Page No.: 42

Procedure:-

1. Fix the black paper on the glass plate by using adhesive.

2. place two razor blades so that their sharp edges are parallel and extremelly close to each other to form a narrow slit in between.

3. cut the small slit in between the sharp edges of blades and place at a suitable distance from a wall or screen of a dark room.

4. Throw a beam of light on the slit by the laser pencil.

5. A diffraction pattern of alternate bright and dark bands is seen on the wall.

conclusion:-

When light waves are incident on a slit or aperture then it bends away (spread) at ~~time~~ comers of slit showing the phenomena of diffraction of light

Teacher's Signature: .....

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Expt. No.: .....

Date : .....

Page No.: 43

Precautions:-

1. Air gaps should not be left between glass plates and black paper.

2. The razor blades should be placed extremly closed as possible.

3. Diffraction pattern should be seen on a wall of a dark room.

4. A point source of monochromatic light like laser torch should be used.

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